

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2025****Course: B. Tech.****Branch: Civil Engineering****Semester: VIII****Subject Code & Name: BTCESS802A Energy Efficiency Acoustics and Daylighting in Building****Max Marks: 60****Date:13/ 01/ 2026****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Tropical environments are mainly characterized by A) Low solar intensity and high wind speed B) High humidity with moderate temperatures C) High solar radiation and high humidity D) Low humidity and low rainfall	CO1	1
2	Relative humidity has maximum adverse impact on comfort when A) Air temperature is low B) Air movement is high C) Skin temperature approaches air temperature D) Mean radiant temperature is low	CO1	1
3	Comfort indices attempt to A) Replace subjective judgement entirely B) Quantify combined environmental parameters C) Define absolute comfort conditions D) Eliminate climatic influence	CO1	1
4	Heat exchange between a building and its environment is governed by	CO2	1

	A) Conduction through envelope only	B) Combined conduction, convection and radiation	C) Radiation as the dominant mode always	D) Convection independent of surface properties		
5	Periodic heat flow in buildings becomes significant mainly due to				CO2	1
	A) Internal heat gains	B) Diurnal variation of outdoor conditions	C) High thermal conductivity of materials	D) Ventilation losses		
6	Decrement factor of a wall is defined as the ratio of				CO2	1
	A) Indoor to outdoor mean temperature	B) Peak heat flow to average heat flow	C) Indoor surface temperature amplitude to outdoor amplitude	D) Heat stored to heat transmitted		
7	The order of the transmission matrix for a single homogeneous layer is				CO3	1
	A) 1×1	B) 2×2	C) 3×3	D) Depends on thickness		
8	Admittance method evaluates heat flow based on				CO3	1
	A) Instantaneous outdoor temperature	B) Average daily temperature	C) Periodic heat exchange at internal surfaces	D) Solar radiation only		
9	Acoustic separation is mainly achieved by				CO4	1
	A) Ventilation design	B) Structural isolation	C) Daylight control	D) Thermal insulation		
10	Orientation of a building primarily influences				CO4	1
	A) Wind pressure coefficients	B) Daylight penetration only	C) Solar heat gain pattern on envelope	D) Structural load distribution		
11	Air-borne noise transmission is primarily controlled by				CO5	1

	A) Increasing surface mass	B) Isolating structural connections	C) Providing resilient mounts	D) Increasing room height		
12	Glare occurs mainly due to				CO5	1
	A) Uniform brightness distribution	B) Excessive luminance contrast	C) Low daylight levels	D) Poor ventilation		
Q. 2	Solve the following.					12
A)	State the factors affecting human comfort.				CO1	6
B)	Discuss the importance of considering local climate in building design.				CO1	6
Q.3	Solve the following.					12
A)	State the effect of solar radiation on buildings.				CO2	6
B)	Explain periodic heat flow and its significance in buildings.				CO2	6
Q. 4	Solve Any Two of the following.					12
A)	Discuss the basic principles of good lighting in buildings.				CO3	6
B)	Write a short note on basic acoustics and noise.				CO3	6
C)	Write a short note on artificial lighting.				CO3	6
Q.5	Solve Any Two of the following.					12
A)	Explain fenestration design for natural ventilation.				CO4	6
B)	What are the types of heat transfer in buildings?				CO4	6
C)	Explain the importance of building orientation in reducing heat gain.				CO4	6
Q. 6	Solve Any Two of the following.					12
A)	Discuss methods to control airborne noise in buildings.				CO5	6
B)	Explain mechanisms of natural ventilation.				CO5	6
C)	Explain outdoor noise in detail.				CO5	6
*** End ***						